

Deep Learning

Introduction

Deep Learning is a subfield of machine learning that uses neural networks with multiple layers to learn complex patterns from data.

Neural Networks

A neural network contains layers of interconnected units. An input layer receives features, hidden layers transform information, and an output layer produces predictions.

Training

During training, a neural network compares predictions with expected results using a loss function. An optimization algorithm such as gradient descent updates model parameters to reduce the loss.

Applications

Deep learning is widely used for image classification, speech recognition, natural language processing, recommendation systems, and computer vision.

Challenges

Deep learning often requires substantial data and computational resources. Models can also be difficult to interpret and may inherit biases present in their training data.